ITEA4 22007
Sa4CPS project

WP3: Perception & Comprehension

D3.2: Technical Enablers for Perception and Comprehension (T3.1-T3.4)

Sa4CPS
ITEA 4 22007

Status: V 1.0

Edited by Murat SAGLAM (Necdet Alpata)

Including contributions from:
Yasemin Eker (Necdet Alpata)
Feyza Nur Genç (Necdet Alpata)
Diogo Sousa (Digitalmente)
Ana Lobo (CEL)
Pedro Faria (ISEP)
Rafael Silva (ISEP)
Dilara Bayar Başlık (ACD)
Edanur Başlık (Inovasyon)
Milad Abdullah (CUNI)
Tomáš Bureš (CUNI)
Petr Hnětynka (CUNI)
Jiří Havlík (IMA)
Hannu Lohi (Natlink)
Topias Nykänen (VTT)
Lander Vanhaverbeke (Vintecc)
Ibrahim Arif (ERARGE)

TABLE OF CONTENTS

TABLE OF FIGURES

Abstract

Secure situational awareness in critical cyber-physical systems (CPS) relies not only on clearly defined requirements but also on the availability of robust technical enablers that operationalize perception and comprehension capabilities. This deliverable identifies and structures the **technical enablers for perception and comprehension** required to address the technological requirements defined in D3.1 across heterogeneous and security-critical CPS domains.

The document describes enablers supporting secure and reliable perception, including heterogeneous sensing, interaction-based data acquisition, real-time data streaming, and trusted integration of physical and cyber data sources. These enablers enable accurate position awareness, continuous environment monitoring, and synchronized physical–digital representations through digital twin technologies.

In addition, D3.2 details comprehension enablers that transform raw and fused data into meaningful situational understanding. These include multi-source sensor and information fusion, AI-based reasoning and inference mechanisms, anomaly detection, and predictive modeling for anticipating abnormal or unsafe system states. Particular attention is given to cross-cutting enablers such as secure data management, low-latency communication, semantic processing, and integrity assurance, which are essential for maintaining trustworthiness and resilience in distributed CPS architectures.

List of Abbreviations

Abbreviation	Explanation
GIS	Geographical Information System
MQTT	Message Queuing Telemetry Transport
AI	Artificial Intelligence
API	Application Programming Interface
AMR	Autonomous Mobile Robot
CSV	Comma-Separated Values
GPS	Global Positioning System
GUI	Graphical User Interface
HSM	Hardware Security Module
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IoT	Internet of Things
KPI	Key Performance Indicator
ML	Machine Learning
M2M	Machine-to-Machine
PHM	Prognostics and Health Management
QoS	Quality of Service
RBAC	Role-Based Access Control
REST	Representational State Transfer
RTSP	Real Time Streaming Protocol
SoH	State of Health
RUL	Remaining Useful Life
TLS	Transport Layer Security
ToF	Time-of-Flight
UC	Use Case
UWB	Ultra-Wideband
VPN	Virtual Private Network
JWT	JSON Web Token
JSON	JavaScript Object Notation
FEM	Front-End Module

1 Introduction

WP3 objectives:

WP3 focuses on the R&D of technical enablers for perception and comprehension to establish reasoning and understanding of the current situation. It is estimated that information sources can be e.g. sensing the environment using sensors, cameras, available radios etc. Interaction between neighbouring computer equipped nodes and service systems in back office/clouds. Smart information processing, sensor fusions, and use of AI/machine learning systems are needed for comprehension of the situation.

Expected results:

Technological requirements for perception and comprehension. Technical enablers for perception and comprehension required in the Sa4CPS concept. Evaluation of the developed enablers for perception and comprehension. Building up physical sensor system consisting of various sensor technology where new data acquisition and data processing mechanisms will be validated and ready for object detection and tracking to be performed in WP4.

Objectives of 3.2:

- Define and document the technical enablers for perception and comprehension

2 Task Descriptions

Task	Description
Task 3.1	**Physical environment sensing** [**Natlink**, IMA, Polar, Digitalmente, CEL, ACD, VTT, Otokar, Enerim] (all UCs) Technical requirements and R&D of the sensing systems, sensors, cameras, radios etc. neighbour discovery and their deployment for exploitation of information from physical environment for situational awareness.
Task 3.2	**Interaction based sensing of information context** [**Otokar**, Natlink, Polar, Digitalmente, ACD, Vintecc, VTT, Enerim] (all UCs) Technical requirements and R&D of the means for Acquisition of information from a other computers nearby, information discovery, and exploitation from cloud environments of different stakeholders.
Task 3.3	**Information & sensor fusion** [**Polar**, Natlink, ISEP, Digitalmente, Natlink, ACD, CEL, INOVASYON, ADLC, Vintecc, VTT, Enerim] (all UCs) Technical requirements and R&D of the technical means for Smart fusion of information from various sources and sensors (sensor fusion). Algorithms and tools for enabling such processes.
Task 3.4	**Modelling and Tools for comprehension** [**INOVASYON**, CUNI, ISEP, CEL, Digitalmente, Natlink, VTT, Polar] (all UCs) Technical requirements and R&D of the technical means for Modelling and tools, e.g. AI/machine learning, for enabling smart comprehension of the specific situations in the practical targeted use cases.

3 Use Case Enablers for Perception and Comprehension

3.1 Connected & Autonomous lightweight logistics for in-factor/out-factory supply chain (UC1)

3.1.1 Perception Enablers

ID	**P_EN_UC1.1**
Name	Low-Latency Hybrid Communication Module
Enabler Type	System / Network Infrastructure
Supported Requirement (D3.1)	REQ_UC1.14
Description	A communication module ensuring seamless handovers between Wi-Fi and 5G networks for mobile assets like forklifts.
Technical Characteristics	Low latency, support for 5G/Wi-Fi handover protocols, high-speed data throughput.
Use Case Scenario Relevance	Ensures that autonomous forklifts maintain constant, real-time contact with control systems without signal drops.
Author	ACD

This enabler ensures continuous connectivity for autonomous forklifts by managing seamless handovers between Wi-Fi and 5G networks.

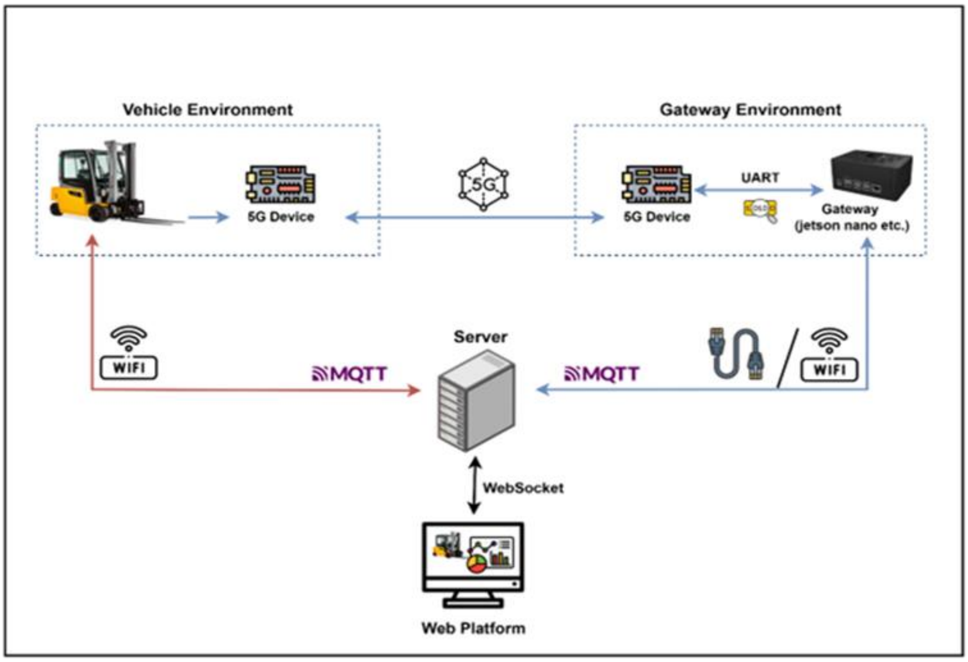


Figure 1. System architecture

5G Mesh Network Architecture: The technical setup utilizes a 5G Mesh Network where development kits are programmed with specific roles: sink, router, and non-router.

Performance Metrics: Field tests with AMR on a 120-meter route demonstrated high reliability; out of approximately 16,000 MQTT messages, only 0.24% experienced delays greater than 500 ms. Most end-to-end transmission delays ranged between 0 and 250 ms.

ID	**P_EN_UC1.2**
Name	Secure Data Sharing via IPFS
Enabler Type	System / Software
Supported Requirement (D3.1)	REQ_UC1.15
Description	A robust infrastructure for the secure storage, processing, and transmission of IoT device data to external networks.
Technical Characteristics	Secure data sharing with IPFS, data transmission to external networks.
Use Case Scenario Relevance	Ensures that logistics data is shared across the supply chain securely and efficiently using decentralized technology.
Author	ACD

This enabler provides a decentralized infrastructure for the secure storage and transmission of the data.

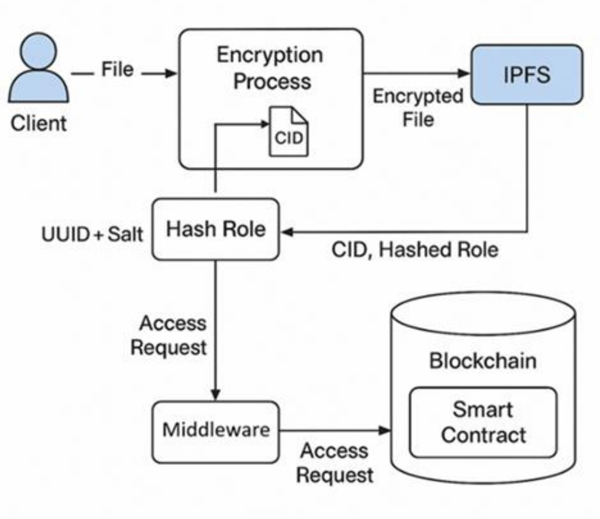


Figure 2. IPFS architecture

Data Pipeline and Storage: Telemetry and sensor data collected via MQTT are first ingested into an ELK Stack (Elasticsearch, Logstash, Kibana) for real-time analysis. Daily accumulated data is exported in CSV format and transferred to a Private IPFS Network running on Docker containers for long-term storage.

Blockchain-Based Proof Layer: To ensure data integrity, the system records the Content Identifier (CID) of every IPFS file onto a private Hyperledger Besu blockchain.

Zero-Trust Verification: Instead of storing the raw data on the blockchain, only the CIDs and verifiable timestamps are recorded. This allows users to validate any retrieved dataset by recalculating its CID and comparing it with the immutable blockchain record, enforcing a zero-trust data verification model.

ID	**P_EN_UC1.3**
Name	Network Component Health Monitor
Enabler Type	Software
Supported Requirement (D3.1)	REQ_UC1.17
Description	A real-time monitoring tool that tracks the status and network status, IoT devices, and Middleware.
Technical Characteristics	Continuous status reporting, health metric evaluation, real-time alert generation.
Use Case Scenario Relevance	Provides the visibility needed to prevent network-related downtime in the autonomous logistics line
Author	ACD

This enabler monitors the status of network components and IoT devices in real time.

- Frontend Architecture: The platform is built on a React-based architecture. It uses WebSocket channels to transmit real-time location, task, and connectivity status from the MQTT infrastructure directly to the web interface.
- Centralized Authentication Service: ACD developed a backend microservice using Node.js and MongoDB (NoSQL). It handles identity management, including email-based verification, cryptographic password hashing, and JSON Web Token (JWT) management.
- Security and Access Control: The platform implements Role-Based Access Control (RBAC) with predefined roles (admin, user, developer). Following a zero-trust model, every request is explicitly verified via a JWT included in the request header.
- Health and PHM Visualization: The platform visualizes critical health metrics, including battery parameters (state of charge, state of health, temperature) and operational indicators, allowing users to distinguish whether data is arriving via Wi-Fi or 5G.

ID	**P_EN_UC1.4**
Name	PHM Big Data Ingestion Pipeline
Enabler Type	Software / Algorithm
Supported Requirement (D3.1)	REQ_UC1.18
Description	A software pipeline designed to collect large volumes of battery and component data from external sources into a central repository.
Technical Characteristics	High-throughput data streaming, time-series data synchronization, support for heterogeneous formats.
Use Case Scenario Relevance	Aggregates the raw data necessary for performing advanced health and life-cycle calculations.
Author	INOVASYON

This enabler is a software pipeline designed to collect large volumes of battery and component data from external sources into a central repository to support advanced health and life-cycle calculations.

Architecture: The system is built as a Node.js-based backend service. It utilizes a hybrid architecture consisting of a REST API for request-response communication and Socket.IO for real-time data streaming to the frontend.

Data Collection (Ingestion): Robot-generated data is collected through MQTT brokers. The backend subscribes to multiple MQTT topics to ingest critical parameters, including:

◦ Battery voltage and current.

◦ Individual cell voltages.

◦ Temperature values.

◦ Task states.

Data Infrastructure: The pipeline uses ElasticSearch for storing and analyzing historical time-series battery data. This infrastructure ensures the system can handle high-throughput data streaming and synchronization of heterogeneous formats.

Real-Time Streaming: Once raw data is processed or computed, it is streamed to the frontend in real time via Socket.IO using a specific event named mqtt_data.

Monitoring and Integrity: The ingestion process is supported by an ELK Stack (Elasticsearch, Logstash, Kibana) for real-time operational monitoring and analysis. This setup provides reliable, scalable, and low-latency delivery of PHM data.

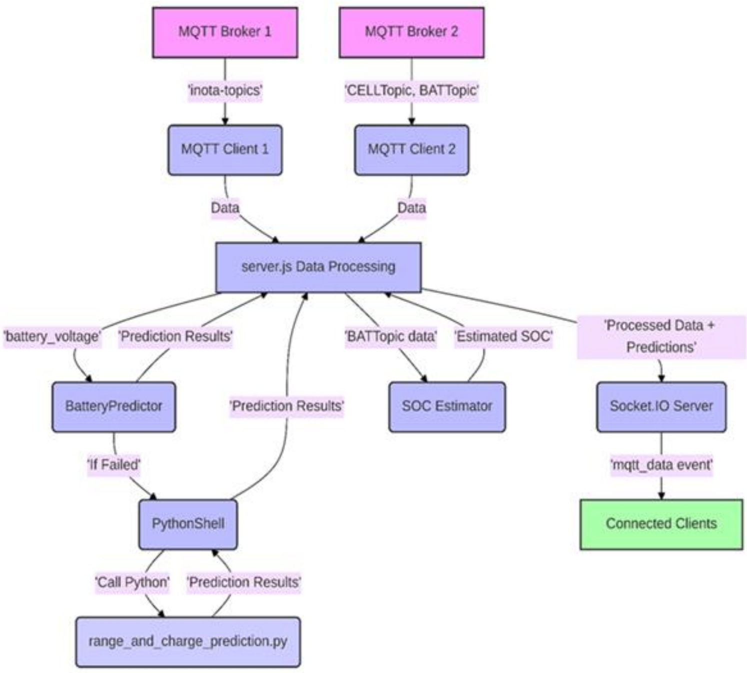


Figure 3. PHM Pipeline

ID	**P_EN_UC1.5**
Name	IoT-based Secure Data Communication Enabler
Enabler Type	Perception - Secure Data Acquisition & Transmission (Hardware)
Supported Requirement (D3.1)	REQ_UC1.34 - IoT-based secure data communication system
Description	This enabler provides secure data collection and communication between edge sensors, IoT gateways, and cloud services within UC1. It ensures that sensor and telemetry data originating from forklifts and logistics assets are transmitted securely, preserving confidentiality, integrity, and authenticity before further processing and comprehension stages.
Technical Characteristics	• Integration of IoT gateways with Hardware Security Modules (HSMs) for cryptographic key management and message signing. • Secure Machine-to-Machine (M2M) communication between sensors, gateways, and cloud services using encrypted channels. • Support for real-time data streaming with minimal latency to enable timely perception. • Device-level trust enforcement through authenticated gateways acting as secure aggregation points. • Compatibility with downstream semantic and AI-based processing pipelines. HSM (left) and Secure IoT Gateway (right) figures: 
Use Case Scenario Relevance	Enables trusted and continuous perception of forklift status, environmental conditions, and logistics events in UC1 by ensuring that raw sensor data is securely delivered to higher-level comprehension and decision-making components.
Author	ERARGE

ID	**P_EN_UC1.6**
Name	IoT-based Asset Identification and Tracking

Enabler Type	System
Supported Requirement (D3.1)	REQ-UC1.3 – Asset Integrity & Authentication REQ-UC1.7 – IoT-Based Monitorization
Description	This enabler provides real-time identification and tracking of automotive parts using IoT beacons and tags. Each asset is uniquely identified and continuously monitored during cross-organisational and in-factory logistics operations.
Technical Characteristics	- IoT beacons / tags attached to automotive parts - Real-time data transmission via IoT gateways - Unique asset IDs linked to backend systems - Low-latency data collection
Use Case Scenario Relevance	Supports UC1 by enabling traceability of parts from external vendors to Otokar factory and during internal transportation with forklifts.
Author	OTOKAR

3.1.2 Comprehension Enablers

ID	C_EN_UC1.1
Name	Battery PHM (Prognostics & Health Management) Tool
Enabler Type	Software / Algorithm
Supported Requirement (D3.1)	REQ_UC1.21, REQ_UC1.22, REQ_UC1.23
Description	An integrated analytical software suite that implements advanced machine learning models and real-time data processing pipelines to manage the health and performance of battery systems.
Technical Characteristics	Fast and reliable processing of high-volume sensor streams, high-accuracy predictive modeling for State of Health (SoH) and Remaining Useful Life (RUL), and automated AI-driven clustering for predictive maintenance.
Use Case Scenario Relevance	Provides the intelligence required for autonomous logistics by identifying potential battery failures and operational anomalies before they disrupt the supply chain.
Author	INOVASYON

ID	C_EN_UC1.2
Name	Optimum Route Planning and Vehicle-to-Vehicle Cooperation Engine
Enabler Type	Software / Algorithm

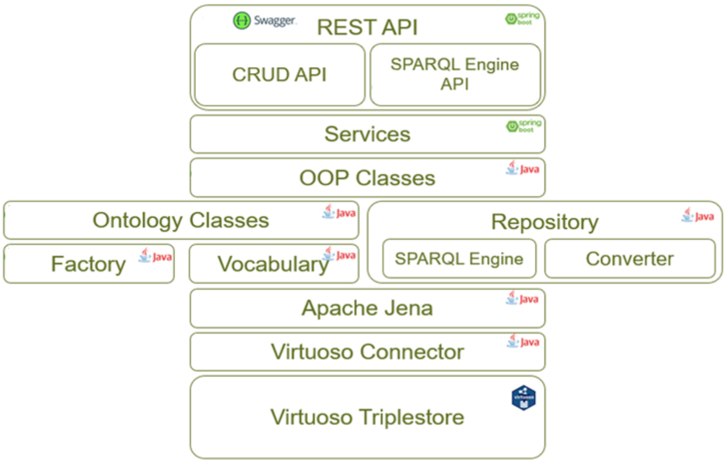
Supported Requirement (D3.1)	REQ_UC1.31, REQ_UC1.32, REQ_UC1.33, REQ_UC1.41, REQ_UC1.42
Description	A decision-support engine fed by IoT-based real-time data streams (GPS location, BLE beacon signals). It provides an integration framework for best-path algorithms selected through literature review (e.g., Dijkstra, A*, and constraint-optimization-based multi-objective route planning). Location, speed, and payload data originating from the Otokar scenario are processed to optimize transportation time, energy consumption, and road safety.
Technical Characteristics	Data Input: Real-time vehicle location data, beacon-based proximity information, and in-vehicle sensor data Algorithms: Multi-criteria route optimization and dynamic re-routing mechanisms System Architecture: Service-oriented architecture with standardized communication interfaces and data buffering Performance: Real-time operation with scalable processing capabilities Security: Secure communication, access control, and data integrity assurance
Use Case Scenario Relevance	Generates dynamic routes based on the real-time locations of autonomous forklifts and transport carts, reducing delays and empty runs. Enables rapid adaptation to Otokar-based scenarios (e.g., human-machine collaboration within a factory), minimizing collision risks by considering real-time road constraints and safety zones.
Author	ALPATA

The Optimum Route Planning and Vehicle-to-Vehicle Cooperation Engine is designed as a modular software enabler supporting real-time decision-making in connected logistics scenarios.

The enabler integrates IoT-based monitoring with semantic data processing to construct a unified operational view of vehicles, routes, and environmental constraints. Routing decisions are computed using classical shortest-path algorithms combined with machine-learning-based prediction models that anticipate congestion and delays.

The system supports continuous route re-evaluation and vehicle-to-vehicle coordination by exchanging route intentions and proximity information, enabling safe and efficient operation in dynamic industrial environments. Validation activities are carried out in simulation-based setups to assess performance, robustness, and safety compliance.

ID	**C_EN_UC1.3**
Name	Semantic Data Processing Enabler for Sensor Fusion and AI Outputs
Enabler Type	Comprehension - Semantic Processing & Reasoning (Software)
Supported Requirement (D3.1)	REQ_UC1.35 - Semantic data processing for sensor fusion and AI algorithm outputs
Description	This enabler enables semantic processing of fused sensor data and AI-generated outputs to transform heterogeneous raw data into meaningful situational knowledge. It supports structured representation, querying, and reasoning over logistics and operational data to enable comprehension of the current and evolving system state in UC1.
Technical Characteristics	• Semantic data management using Virtuoso Triplestore for storing enriched operational and sensor data. • Integration of data streaming and processing components (e.g., Kafka-based pipelines) for handling AI outputs and sensor fusion results. • SPARQL-based querying and filtering mechanisms to support situational reasoning and event detection. • Structured data exchange using JSON-based representations for interoperability with AI/ML components. • Support for near real-time semantic reasoning over digital twin-aligned data models. Preliminary semantic backend architecture figure:

	
Use Case Scenario Relevance	Enables comprehension of forklift operations and logistics workflows in UC1 by allowing AI algorithms and reasoning mechanisms to operate on semantically enriched, fused sensor data, supporting anomaly detection, prediction, and situational awareness.
Author	ERARGE

ID	**C_EN_UC1.4**
Name	Digital Twin-based Situational Awareness
Enabler Type	System
Supported Requirement (D3.1)	REQ-UC1.3 – Real-time sensing and identification of physical assets
Description	This enabler creates real-time digital twins of physical assets, vehicles, and logistics processes. It enables operators to understand the current system status and interactions between assets, people, and vehicles.
Technical Characteristics	- Real-time synchronization with physical assets - Integration with IoT sensor data - Visual representation via dashboards and 3D views - Support for historical and live data
Use Case Scenario Relevance	Used in UC1 to visualize forklift operations, asset movements, and safety-critical situations inside the Otokar factory.
Author	OTOKAR

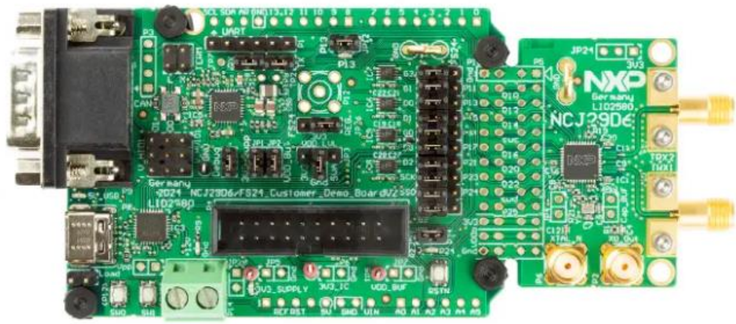
ID	**P&C_EN_UC1.6**
Name	Real-time Anomaly Detection and Alerting
Enabler Type	System
Supported Requirement (D3.1)	REQ-UC1.9 – Situational Awareness & Alerts

Description	This enabler detects abnormal or risky situations in real time and generates alerts to operators. It combines sensor data and analytics to support early awareness and faster response.
Technical Characteristics	- Real-time event processing - Rule-based and predictive analytics - Alert generation via dashboard, email, or visual indicators - Configurable thresholds (warning / danger)
Use Case Scenario Relevance	Applied in UC1 for safety scenarios such as forklift–human proximity detection and abnormal logistics events.
Author	OTOKAR

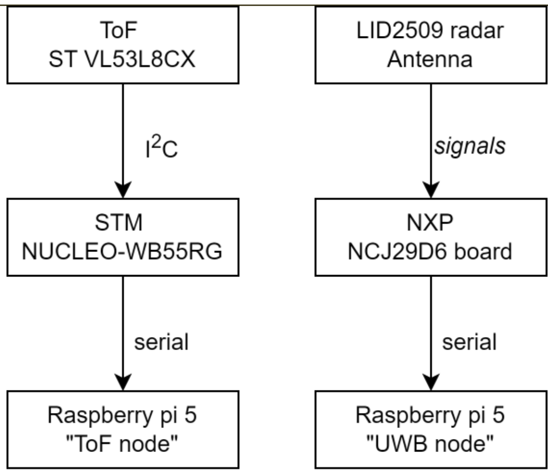
3.2 People and Machine position awareness in production or logistics area (UC2)

3.2.1 Perception Enablers

ID	P_EN_UC2.1
Name	ToF camera sensor and radar for object position
Enabler Type	Hardware
Supported Requirement (D3.1)	REQ_UC2.6
Description	The selection of hardware components has been driven by the main goal of the project: to enable a safe and adaptive collaboration between human operators and robotic systems. The system integrates multiple sensing modalities positioned strategically around the workplace. Each component has been selected according to parameters such as range, accuracy, robustness, and ease of integration.
Technical Characteristics	The Time-of-Flight (ToF) sensor used in the demonstrator is based on the ST VL53L8CX module integrated by SATEL. This latest-generation sensor provides real-time depth measurements in an 8×8 or 4×4 grids. • Resolutions: 8x8 or 4×4 (64 or 16 points per frame) • Field of View: 65° × 65° • Range: up to 4 m (typ. 2.5 m indoor) • Frame rate: up to 60 Hz • Accuracy: ±5–15 mm depending on range • Interface: I2C This sensor has been selected for its high accuracy, wide field of view, and compact form factor, which enables safe and non-intrusive human detection. The ToF data

	are continuously streamed into the processing unit and visualized in a dedicated GUI for real-time safety zone evaluation.  UWB Passive Radar Module: a) NXP NCJ29D6 UWB demo board. b) LID2509 radar Antenna.  The NXP NCJ29D6 ultra-wideband (UWB) radar module, equipped with the LID2509 antenna array, enables precise motion and distance sensing even in challenging environments. Unlike optical sensors, it operates reliably through smoke, dust, or partial obstructions. In the proposed setup, the UWB unit is positioned laterally to monitor human motion around the workstation and assist in safety-zone validation. • Frequency range: 6.5–8.0 GHz (UWB band) • Range resolution: ≈ 5–10mm • Maximum range: up to 5m • Frame rate: up to 50 Hz • Antenna: LID2509 dual-polarized radar array
Use Case Scenario Relevance	UC2_1, UC2_2, UC2_3, UC2_4, UC2_5, UC2_6
Author	IMA

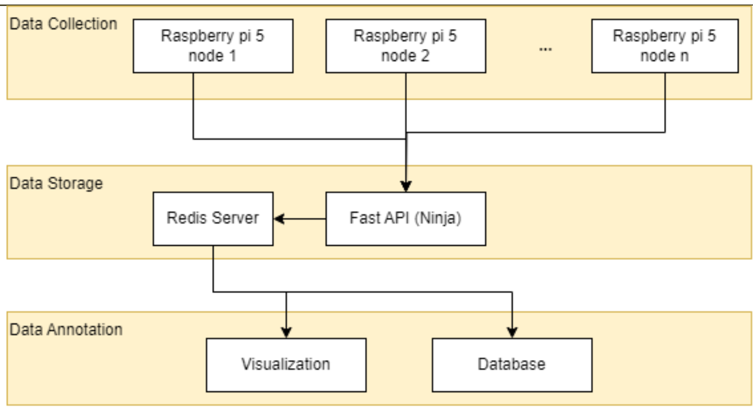
ID	**P_EN_UC2.2**
Name	Data collection from sensors
Enabler Type	Software, service
Supported Requirement (D3.1)	REQ_UC2.6
Description	Data collection software handles the low-level data collection from ToF and UWB radar sensors through dedicated microcontroller units. The STM and NXP processors interface directly with the sensors, managing I2C and signal communication protocols, respectively. These processors are responsible for acquiring raw sensor data at high frame rates and preparing it for upstream processing. The boards are then managed by a Raspberry Pi node. This ensures system stability, scalability, and real-time operation for the CPS safety framework.
Technical Characteristics	The data collection layer consists of two main processing units: STM Processor: • Interfaces with ToF sensor via I2C • Handles sensor initialization and configuration • Buffers raw depth data for transmission to Raspberry Pi • Supports runtime configuration changes (resolution, frame rate) NXP Processor: • Interfaces with UWB radar • Manages antenna array configuration • Buffers radar data for upstream transmission • Supports dynamic parameter adjustment Both processors are remotely controlled and flashed by a Raspberry Pi controller, which serves as the central coordination unit for sensor configuration and data retrieval.

	 The software system has been fully implemented in Python and is stored in the internal IMA GitLab repository. A dedicated graphical interface has been developed for the ToF sensor, enabling real-time visualization, data logging, and safety state detection. All components are executed inside an isolated virtual environment.
Use Case Scenario Relevance	UC2_1, UC2_2, UC2_3, UC2_4, UC2_5, UC2_6
Author	IMA

ID	**P_EN_UC2.3**
Name	Communication interfaces
Enabler Type	Hardware, Software
Supported Requirement (D3.1)	REQ_UC2.3
Description	The Communication Enabler ensures that sensor information serving as the primary infrastructure for high-speed data exchange between system nodes. It links edge devices (sensors and processors) to local gateways and the backend cloud infrastructure. The Raspberry Pi acts as a central coordination bridge, controlling the STM and NXP processors, managing firmware updates, requesting sensor data, and adjusting runtime configurations. The system controls data flow, manages congestion, and implements packet priority through Quality of Service mechanisms. It aggregates data from multiple sensors and uploads it to the server via network protocols, ensuring synchronized data flow through a Redis queue system for high-throughput processing.

	To maintain security while ensuring fast data transfer, the Raspberry Pi nodes and the data collection server operate within a dedicated VPN infrastructure, independent of IMA or CUNI network dependencies. This architecture ensures project components remain both secure and independent while maintaining high-throughput communication. The communication layer abstracts the complexity of networking, so other system components can focus on their core logic.
Technical Characteristics	Hardware Characteristics: • Raspberry Pi as central gateway and coordination node • Interfaces with STM and NXP processors via GPIO, I2C, SPI, and UART • Network connectivity via Ethernet and/or WiFi (802.11ac/ax) • Multi-band wireless support (2.4 GHz, 5 GHz) for balancing range and throughput • RF Front-End capabilities for reliable wireless transmission • Support for MIMO configurations to increase data throughput • Hardware-level packet prioritization and QoS support Software Characteristics: • Low-level firmware for processor communication and control • Device drivers integrating hardware with Linux network stack • Implementation of communication protocols (TCP/IP, HTTP, WebSocket) • IEEE 802.11 protocol suite for wireless connectivity • Redis queue for asynchronous, high-speed data buffering • Quality of Service (QoS) management for prioritizing critical sensor data • Congestion control and flow management algorithms • Security frameworks for encrypted data transmission • VPN infrastructure for secure communication channels • Independent network architecture (outside IMA/CUNI dependencies)

	- End-to-end encryption for data in transit - Authentication and authorization mechanisms - Secure tunnel establishment and maintenance - Network isolation ensuring component independence Raspberry Pi Control System: - Flashes and configures STM and NXP processors - Issues data requests to sensor processors - Dynamically adjusts sensor parameters (resolution, FPS, gain, etc.) - Aggregates multi-modal sensor data streams - Implements local data buffering and preprocessing - Communicates with FastAPI server over HTTP/WebSocket via VPN - Utilizes Redis queue for asynchronous data upload - Ensures data synchronization across multiple sensing modalities - Handles network reconnection and error recovery - Supports remote configuration updates - Time-critical task management (beacon timing, ACK handling) - Operates within secure VPN tunnel for all server communications - Independent network operation outside organizational dependencies The Redis queue implementation matches the collection speed with upload throughput, preventing data loss during high-rate sensor acquisition. The communication layer abstracts networking complexity, enabling seamless data flow from edge sensors to cloud infrastructure. All Raspberry Pi nodes and the data collection server operate within a dedicated VPN infrastructure, ensuring secure, encrypted communication independent of IMA or CUNI network constraints. This architecture provides high-speed data transfer while maintaining security and system independence.

	
Use Case Scenario Relevance	UC2_1, UC2_2, UC2_3, UC2_4, UC2_5, UC2_6
Author	IMA, CUNI

3.2.2 Comprehension Enablers

ID	**C_EN_UC2.4**
Name	Data annotation and processing
Enabler Type	Service
Supported Requirement (D3.1)	REQ_UC2.6
Description	This enabler provides the backend server infrastructure for receiving, processing, storing, and visualizing sensor data. The FastAPI-based server receives data streams from Raspberry Pi units, stores them temporarily, and provides two main functionalities: real-time visualization for monitoring and analysis, and data recording for ML dataset creation. The visualization interface assists with manual annotation tasks, enabling the creation of labeled datasets for training object detection and localization models. A Redis queue system ensures high-throughput data ingestion, matching the sensor collection rates.
Technical Characteristics	FastAPI Server (Ninja framework): • RESTful API endpoints for data ingestion • WebSocket support for real-time data streaming • Redis queue integration for high-speed data buffering • Temporary storage for recent sensor data • Real-time visualization endpoints for ToF and UWB data • Data retrieval API for playback and analysis • Record conversion pipeline (sensor data → database records)

	- Database integration for persistent storage - Annotation interface support - Multi-modal data synchronization - Scalable architecture for multiple sensor units Data Processing Pipeline: - Ingestion: Receives sensor data via Redis queue - Temporary storage: In-memory or cache-based buffering - Visualization: Real-time rendering of sensor readings - Recording: Conversion to structured database records - Annotation support: Interface for manual labeling - Dataset creation: Export functionality for ML training
Use Case Scenario Relevance	UC2_1, UC2_2, UC2_3, UC2_4, UC2_5, UC2_6
Author	CUNI

ID	**C_EN_UC2.5**
Name	ML-based training and inference
Enabler Type	Software
Supported Requirement (D3.1)	REQ_UC2.1, REQ_UC2.7, REQ_UC2.8, REQ_UC2.9
Description	This enabler encompasses the machine learning pipeline for training object detection and localization models from collected sensor data. The system trains models once sufficient annotated data is accumulated from the ToF and UWB sensors. The ML models learn to identify and localize objects in the workspace using multi-modal sensor inputs. As models mature through iterative training, they transition from data collection mode to inference mode, providing real-time object detection capabilities for the safety monitoring system.
Technical Characteristics	ML Training Pipeline: - Dataset creation from annotated sensor records - Multi-modal data fusion (ToF + UWB) - Support for object detection and localization tasks - Iterative training as new data becomes available - Model versioning and performance tracking - Validation on held-out test sets - Hyperparameter optimization

	• Training frameworks: PyTorch/TensorFlow Model Architecture: • Input: ToF depth maps (8×8 or 4×4) + UWB radar data • Output: Object positions and classifications • Real-time inference capability (<100ms latency) • Lightweight models for edge deployment • Support for model updates without system downtime Inference System: • Real-time object detection and localization • Integration with sensor data pipeline • Low-latency prediction (<100ms) • Confidence scoring and uncertainty estimation • Model performance monitoring • Fallback to traditional methods if confidence is low
Use Case Scenario Relevance	UC2_1, UC2_2, UC2_3, UC2_4, UC2_5, UC2_6
Author	CUNI

ID	**C_EN_UC2.6**
Name	Object Identification and localization
Enabler Type	Service
Supported Requirement (D3.1)	REQ_UC2.14, REQ_UC2.15, REQ_UC2.16
Description	This enabler represents the production-ready object identification and localization system that utilizes mature ML models to provide real-time detection capabilities. Once trained models achieve sufficient accuracy, they are deployed to continuously monitor the workspace, identifying and localizing objects from the ToF and UWB sensor streams. This system enables safety-critical applications by providing accurate spatial awareness of objects and humans in the collaborative workspace, supporting collision avoidance and adaptive robot behavior.
Technical Characteristics	Object Detection System: • Real-time processing of ToF depth maps and UWB radar data • Multi-object detection and tracking

	• 3D position estimation (x, y, z coordinates) • Object classification (human, tool, obstacle, etc.) • Tracking persistence across frames • Motion prediction and trajectory estimation Performance Characteristics: • Latency: <100ms end-to-end (sensor to detection) • Detection range: Up to 4m (ToF) and 5m (UWB) • Position accuracy: ±5-15mm (ToF), ±5-10mm (UWB) • Frame rate: Up to 50-60 Hz • Multi-target tracking: Up to 10 simultaneous objects Integration: • Safety zone monitoring and violation detection • Real-time alerts for collision risk • Robot control system integration • Visualization dashboard for operators • Logging and event recording • API endpoints for external system integration
Use Case Scenario Relevance	UC2_1, UC2_2, UC2_3, UC2_4, UC2_5, UC2_6
Author	CUNI

3.3 Map at the centre of situational awareness in addition to real time video (UC3)

3.3.1 Perception Enablers

ID	**P_EN_UC3.1**
Name	IP camera source
Enabler Type	System / Software
Supported Requirement (D3.1)	REQ_UC3.8, REQ_UC3.14
Description	Online camera capable of steady video streaming.

Technical Characteristics	Reasonably high placement (10m or higher) for a good bird's eye view to allow detection algorithms to operate at peak accuracy. A camera can be a physical device or a simulated 'camera' for example, from Unity engine's view that is converted into an IP camera source. Field of view, and viewing angle do not require strict limits, but their values need to be accurately known to ensure valid position calculation by the AI algorithm doing the image analysis.
Use Case Scenario Relevance	Allows surveillance of secured areas.
Author	Modirum Gespi

3.3.2 Comprehension Enablers

ID	**C_EN_UC3.1**
Name	Network streaming service for online cameras
Enabler Type	Software
Supported Requirement (D3.1)	REQ_UC3.9
Description	Secure perception layer for receiving live video feed from multiple online sources.
Technical Characteristics	The Network Streaming Service supports both RTSP streams and the ONVIF standard, enabling connectivity to multiple video sources of different types. The service is not public and requires the system's users to manually add the sources ensuring that malicious operators can't inject bad data into the system.
Use Case Scenario Relevance	Allows maximum coverage in the secured area's surveillance.
Author	Modirum Gespi

3.4 Critical electricity distribution grid operation in energy communities (UC 4)

3.4.1 Perception Enablers

ID	**P_EN_UC4.1**
Name	Autonomous detection and mitigation of voltage issues
Enabler Type	Software

Supported Requirement (D3.1)	REQ_UC4.10
Description	Focus on the situational awareness frameworks capable of distinguishing between nominal grid states and emergency conditions
Technical Characteristics	Anomaly detection algorithms and monitoring systems
Use Case Scenario Relevance	Disaster prevention mitigation and discovery of anomalies in the distribution network
Author	ISEP

ID	**P_EN_UC4.2**
Name	Secure Data Ingestion & Validation Layer
Enabler Type	System / Software
Supported Requirement (D3.1)	REQ_UC4.3, REQ_UC4.4, REQ_UC4.10
Description	Secure perception layer responsible for ingesting, validating, and authenticating IoT and metering data originating from distributed energy resources, consumer premises, and grid assets.
Technical Characteristics	Message integrity verification and timestamp validation Detection of missing, duplicated, or anomalous telemetry Edge/fog deployment to minimize data exposure
Use Case Scenario Relevance	Ensures that grid and consumption data used for situational awareness and flexibility activation is trustworthy, tamper-resistant, and compliant with critical infrastructure security expectations.
Author	Digitalmente

ID	**P_EN_UC4.3**
Name	Consent-Aware Energy Data Perception Module
Enabler Type	Software
Supported Requirement (D3.1)	REQ_UC4.6, REQ_UC4.7, REQ_UC4.8
Description	Perception module that filters and contextualizes incoming energy data based on explicit consumer consent and role-based access rules, ensuring GDPR-compliant data acquisition for energy community operations.
Technical Characteristics	Consent registry linked to consumer identities Attribute-level data filtering Role-based data exposure (consumer / CEL operator / DSO) Audit logging of data access and usage

Use Case Scenario Relevance	Allows CEL to perceive only authorized data while maintaining consumer trust and regulatory compliance.
Author	Digitalmente, CEL

3.4.2 Comprehension Enablers

ID	**C_EN_UC4.1**
Name	Digital twin for electricity distribution network with distributed generation and flexible load
Enabler Type	Software
Supported Requirement (D3.1)	REQ_UC4.1, REQ_UC4.2
Description	Digital twin for the electricity distribution network serves as the foundational analytical engine, utilizing state estimation algorithms and historical baseline datasets. It synthesizes virtual sensor data to maintain operational continuity in the event of telemetry loss
Technical Characteristics	State prediction
Use Case Scenario Relevance	Ensure that, in case of failure of communication, data loss is minimized
Author	ISEP

ID	**C_EN_UC4.2**
Name	Dashboard for energy community management
Enabler Type	Software
Supported Requirement (D3.1)	REQ_UC4.7, REQ_UC4.13, REQ_UC4.14
Description	UI and a series of REST API calls designed to handle the lifecycle of community participants and allow operators to monitor production and consumption trends at both the prosumer and community levels
Technical Characteristics	Permissioned data for consumers, automated reporting and tracking of KPI analytics
Use Case Scenario Relevance	Providing transparency for the users, encouraging participation in the community. Also, enables management by the DSO
Author	ISEP

ID	**C_EN_UC4.3**
Name	Activation of flexible resources
Enabler Type	Software
Supported Requirement (D3.1)	REQ_UC4.12

Description	Real-time actuation through the activation of flexible resources, specifically targeting battery energy storage systems and electric vehicle integration
Technical Characteristics	Dynamic response constraints to ensure requirements are met and robust communication with deployed assets
Use Case Scenario Relevance	Enables the actuation in cases of emergencies, voltage issues or overloads.
Author	ISEP

ID	**C_EN_UC4.4**
Name	Energy Community Situational Comprehension Engine
Enabler Type	Software
Supported Requirement (D3.1)	REQ_UC4.1, REQ_UC4.11, REQ_UC4.13, REQ_UC4.17
Description	Transforms perceived grid, consumption, and flexibility data into actionable situational understanding for energy community operators. It provides aggregated views, trend analysis, and early warnings for abnormal or inefficient states.
Technical Characteristics	Aggregation of consumption, production, and flexibility signals KPI computation (self-consumption, peak reduction, emissions) Detection of abnormal community behavior Visualization tailored to CEL operational workflows Support for scenario comparison and historical analysis
Use Case Scenario Relevance	Enables CEL operators to understand the current and projected state of the energy community and act proactively.
Author	Digitalmente, CEL, ISEP

ID	**C_EN_UC4.5**
Name	Trust & Data Quality Assessment Engine
Enabler Type	Software
Supported Requirement (D3.1)	REQ_UC4.1, REQ_UC4.3
Description	Reasoning component that continuously evaluates the trustworthiness, freshness, and completeness of energy data streams and digital twin states, assigning confidence levels to situational outputs.

Technical Characteristics	Data quality scoring (freshness, completeness, consistency) Confidence indicators for operators Integration with digital twin state estimation Fallback logic in case of telemetry loss Explainable reasoning outputs
Use Case Scenario Relevance	Prevents incorrect decisions caused by missing or unreliable data and strengthens operator confidence in automated recommendations.
Author	Digitalmente

3.5 Multirouting secure awareness system for hunting and outdoors (UC 7)

3.5.1 Perception and Comprehension Enablers

ID	**P_EN_UC7.1**
Name	Multicommunicator / Sync Hub
Enabler Type	Device
Supported Requirement (D3.1)	REQ_UC7.10, REQ_UC7.11, REQ_UC7.12, REQ_UC7.13, REQ_UC7.14, REQ_UC7.15, REQ_UC7.2, REQ_UC7.20, REQ_UC7.21, REQ_UC7.3, REQ_UC7.4, REQ_UC7.5, REQ_UC7.9
Description	The Multicommunicator is a versatile communication hub that bridges long-range and short-range wireless links to reliably route device and phone data even in areas with limited or no network coverage.
Technical Characteristics	The Multicommunicator is a waterproof, field-ready communication hub powered by dual 18650 batteries and built around three main radios: an LTE Cat M1 / Cat1-bis modem, a WiFi-Bluetooth (5.4) controller, and a LoRa subsystem for long-range low-power connectivity. It supports both long-range and short-range wireless links, encrypted beacon protocols, and efficient data routing, enabling reliable multi-technology communication even in remote areas with limited network coverage. The Multicommunicator improves mobile phone coverage by letting the phone use its high-sensitivity LTE Cat M1 / Cat 1-bis modem, stronger antenna, and roaming SIM, effectively doubling the practical coverage range compared to a phone alone. It acts as an external modem that the phone connects to via Bluetooth or WiFi, ensuring stable connectivity even in weak-signal areas. It has 5 operation modes Modes used in Natlink Use case

	1. Full Operation Mode – Routes IP traffic from LTE Cat M1 / Cat 1-bis to the mobile phone via WiFi and forwards LoRa data from tracking devices. 2. LoRa Router Mode – Acts as a standalone LoRa-to-LTE router with WiFi disabled for extended battery life. 3. LoRa Receiver-Only Mode – Sends LoRa data directly to the mobile phone while LTE and WiFi remain off to save power. 4. Power Bank Mode – Functions as a power source for the user's mobile phone using its dual 18650 batteries. Mode used in Polar Use case 5. Sync Hub Mode – Operates as a local short-range Bluetooth/BLE Coded PHY receiver / router, relaying device data to server and back, maybe controlled by local tablet or phone UI 6. All Polar wearable devices with BLE-SDK interface can be used with Sync Hub 
Use Case Scenario Relevance	UC7
Author	Natlink, Polar

ID	**P_EN_UC7.1**
Name	LoRa Dog tracking devices
Enabler Type	Device
Supported Requirement (D3.1)	REQ_UC7.15
Description	LoRa-based dog tracking prototypes 3 different types. Devices are designed for long-range, and low power positioning in areas with poor mobile coverage. The prototypes undergo iterative field testing, with new PCB revisions, radio-performance improvements, and integration into the Multicommunicator ecosystem, enabling direct LoRa point-to-point tracking, extended range (1–5 km), and future options such as hybrid LoRa/cellular and Bluetooth Long Range operation.
Technical Characteristics	The first set of the LoRa tracking devices which already have been implemented were: The Artemis-based LoRa prototype offers the longest range at 3–5 km, thanks to its larger physical size and room for a more efficient antenna structure. The Luna-based version provides a 2–4 km range, balancing size, RF performance and updated mechanics. The Bark-based version is the most compact, resulting in shorter 1–2.5 km range due to its smaller enclosure and more limited antenna geometry—the main differences between all versions being device size and antenna performance, which directly determine achievable LoRa range. They were based on Microchip LoRa product family WLR089U0. Now it is planned to do next generation with Tracker Bark mechanics and Modified Tracker Luna mechanics which are then based on STM32WLE5 + external FEM (500 mW). With this model is then tested full functionality including high level ciphering, over the air updates, two way traffic, congestion handling with timeslot based allocation etc.
Use Case Scenario Relevance	UC7
Author	Natlink

3.6 Secure, trustworthy and reliable drone operation control system (UC 8)

3.6.1 Perception Enablers

ID	**P_EN_UC8.1**
Name	Secure Multi-Source Perception Ingress Layer

Enabler Type	Software / System
Supported Requirement (D3.1)	REQ_UC8.14, REQ_UC8.15, REQ_UC8.16, REQ_UC8.17, REQ_UC8.22, REQ_UC8.25
Description	A secure, protocol-agnostic ingestion and normalization layer that collects information from onboard drone sensors, ground-based sensors, and third-party environmental services, verifies their integrity, and converts them into a unified internal data representation.
Technical Characteristics	Implemented as distributed ingress services running on cloud services. Connectors for: - MQTT telemetry (aircraft, health, onboard sensors), - HTTPS APIs (weather, METAR, aviation data, KP index), - Ground sensors (cameras, radar, weather stations via REST/MQTT). Integrity verification: - Temporal plausibility checks, - Source authentication, - Frequency monitoring, - Statistical anomaly detection on streams. Secure transport (TLS, OAuth2, mutual authentication where applicable). Authenticated and validated information is published to internal message queue topics. The services are designed to scale horizontally per sensor class or provider.
Use Case Scenario Relevance	UC8
Author	ADLC

ID	**P_EN_UC8.2**
Name	Cyber-Physical Sensor Fusion Pipeline
Enabler Type	Software / System
Supported Requirement (D3.1)	REQ_UC8.18, REQ_UC8.20, REQ_UC8.24, REQ_UC8.25
Description	A real-time fusion engine that correlates heterogeneous sensor streams (physical, cyber, environmental, and network) into a coherent multi-domain operational perception layer.

Technical Characteristics	Implemented using per-flight worker processing contexts. The worker listens to all message queue topics, filters information relevant to the flight, and fuses it into a coherent operational state. It performs: - Temporal alignment, - Spatial correlation, - Cross-domain consistency checks.
Use Case Scenario Relevance	UC8
Author	ADLC

3.6.2 Comprehension Enablers

ID	**C_EN_UC8.1**
Name	Situational Comprehension Engine
Enabler Type	Software / System
Supported Requirement (D3.1)	REQ_UC8.19, REQ_UC8.23, REQ_UC8.24
Description	A predictive reasoning layer that interprets fused operational state and identifies anomalies using rule-based logic combined with statistical techniques.
Technical Characteristics	This is implemented inside the worker processing layer. It will use: • Time-series models for trend detection, • Autoregressive forecasting for energy, link quality, deviation risks, • Pattern classification for abnormal situations. Historical data will be used for better trend understanding and detection.
Use Case Scenario Relevance	UC8
Author	ADLC

3.7 Situational awareness and secure management of digital twin assets (UC 9)

3.7.1 Perception Enablers

ID	**P_EN_UC9.1**
Name	Capture platform (Collecting)
Enabler Type	Software

Supported Requirement (D3.1)	REQ_UC9.5
Description	The Capture platform collects information on the CPS
Technical Characteristics	
Use Case Scenario Relevance	This is required to allow situational awareness
Author	Vintecc

3.7.2 Comprehension Enablers

ID	**C_EN_UC9.1**
Name	Capture platform (contextualizing)
Enabler Type	Software
Supported Requirement (D3.1)	REQ_UC9.6
Description	The Capture platform allows contextualizing data, so meaningless numbers become useful information that can be understood and acted upon.
Technical Characteristics	
Use Case Scenario Relevance	The system can infer meaningful insights from raw data and accurately assess whether a proposed action aligns with safe operational conditions.
Author	Vintecc

3.8 Technical enabler from the project story: 3D situational awareness map of Oulu port

3.8.1 Perception enablers

ID	**P_EN_UC3.1**
Name	3D Sitaw Map
Enabler Type	System
Supported Requirement (D3.1)	REQ_UC3.6, REQ_UC3.7 (REQ_UC3.11, REQ_UC3.12)
Description	A georeferenced 3D virtual map representing a real-world port area, providing operators with immediate spatial awareness of terrain, buildings, and entities. The map acts as the primary visual layer through which positional data is perceived.
Technical Characteristics	• Implemented in Unity game engine • Non-GIS-based georeferencing solution with 1:1 meter-to-unit scaling • Accurate terrain elevation and building placement

	• Layer-based visualization enabling selective display of map elements • Modular and extensible scene architecture
Use Case Scenario Relevance	Supports rapid perception of spatial relationships between assets, infrastructure, and terrain in complex operational environments, forming the foundation for location-aware monitoring and security assessment.
Author	VTT

3.8.2 Comprehension enablers

ID	**C_EN_UC3.1**
Name	3D Sitaw Map
Enabler Type	System
Supported Requirement (D3.1)	REQ_UC3.6, REQ_UC3.7 (REQ_UC3.11, REQ_UC3.12)
Description	Builds on the 3D map by enabling interpretation of spatial constraints and operational context through geofencing, layered information control, and clear visualization of object positions relative to restricted or allowed areas.
Technical Characteristics	• Polygon-based geofencing tied to georeferenced coordinates • Visual indication of area constraints and object compliance • Toggleable information layers to reduce visual clutter • Consistent coordinate framework shared across map features
Use Case Scenario Relevance	Enables operators to understand whether entities are behaving as expected within defined spatial rules, supporting decision-making related to access control, safety zones, and situational compliance in cyber-physical environments.
Author	VTT

3.8.3 3D situational awareness map of Oulu port

The developed 3D virtual environment represents a georeferenced, interactive representation of the Port of Oulu, created to support situational awareness research within the Sa4CPS project. The system functions as a lightweight digital twin or digital shadow of the port area, where terrain, buildings, and operational entities are placed according to real-world coordinates. Rather than relying on a full GIS (Geographical Information System) stack, the environment prioritizes a controllable and extensible implementation that supports simulation, visualization, and system-level experimentation.

The virtual environment will be implemented using the Unity game engine and integrated with the VTT-developed SITU architecture, acting as a visual front-end for cyber-physical data. It provides a spatially grounded view of object locations, movements, and security-related metadata originating from backend services. Development was guided by a narrative scenario involving a lumber shipment arriving by truck at the Port of Oulu, which served as a realistic reference for validating spatial behavior and interaction flows.

The primary role of the system is to enhance location-based perception and comprehension within the situational awareness pipeline. By anchoring data to a realistic 3D map, the environment supports intuitive reasoning regarding distances, spatial relationships, and operational constraints. Within Sa4CPS, the simulator serves as:

- A sandbox for visualizing and validating data flows and system interactions
- A demonstration platform for scenario-based research and stakeholder communication
- A foundation for hybrid simulation combining live, recorded, and simulated data

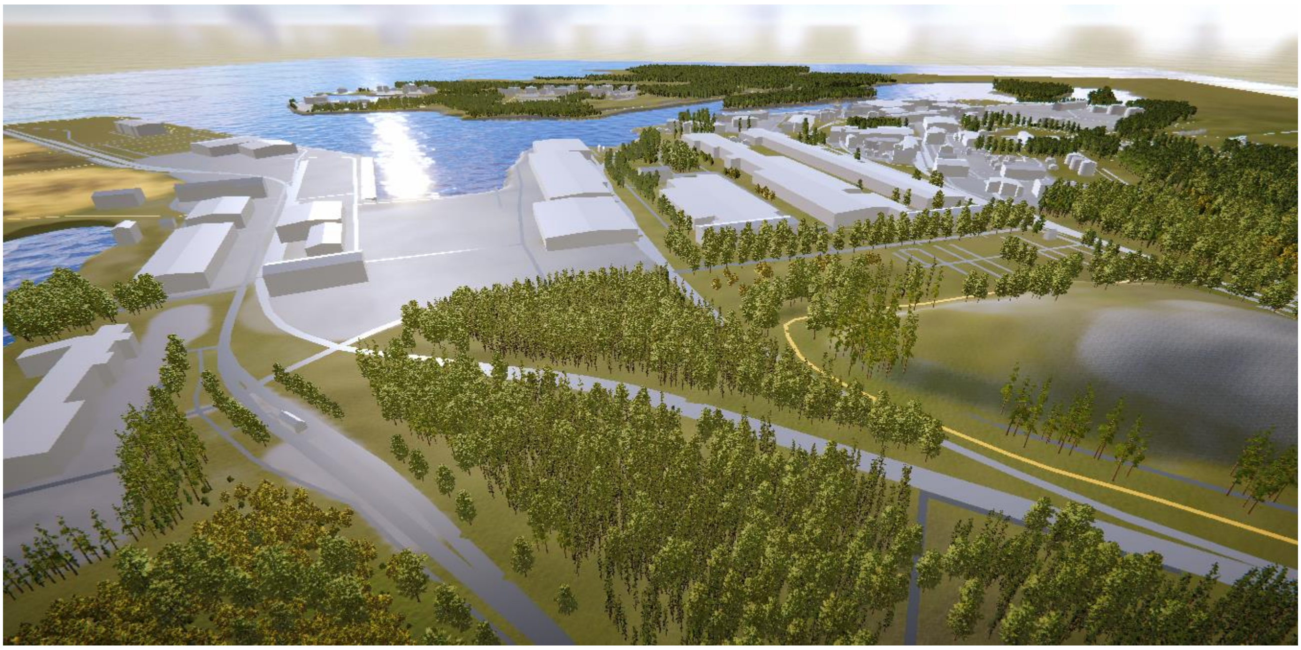
At the technical level, the system uses a lightweight, non-GIS georeferencing solution to transform latitude-longitude coordinates into Unity world space. Terrain and building models are positioned accordingly, forming the spatial backbone for all higher-level functions. Dynamic entities, such as vehicles and assets, can be instantiated and navigated using waypoint-based movement, while spatial constraints are enforced through geofencing mechanisms. Evaluation results indicate that the achieved accuracy (~0.45meters) is acceptable for the current development stage, with clear potential for refinement should system requirements evolve.

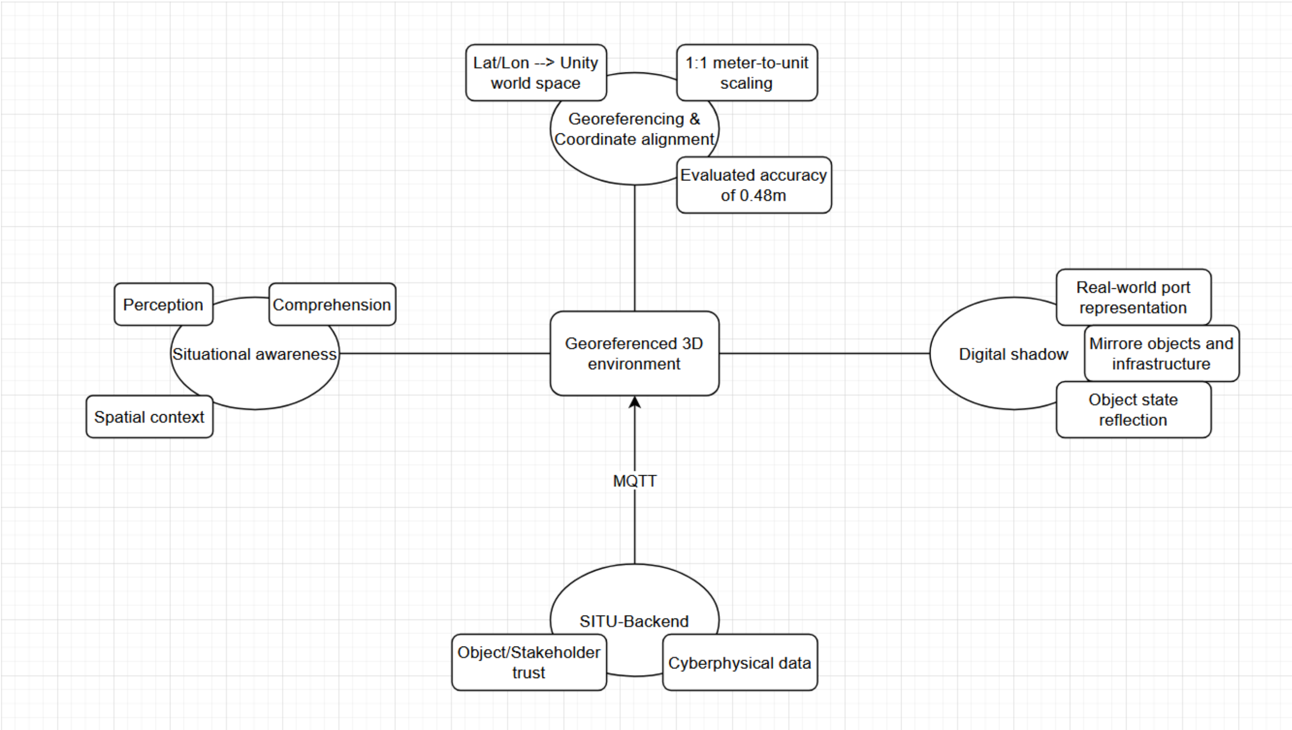
Key functional capabilities include:

- Georeferenced terrain and building placement with real-world scale consistency (0,45m accuracy)
- Visualization of dynamic objects driven by backend location and state updates (primarily via MQTT (Message Queuing Telemetry Transport) via the SITU-system)
- Support for geofencing and route-based navigation tied to geographic coordinates

The user interface was intentionally kept minimalistic, emphasizing visibility of the environment while providing essential contextual information through simple icons and textual indicators. Standard mouse-and-keyboard interaction and familiar camera controls ensure usability without extensive training.

Overall, the virtual environment seeks to contribute to Sa4CPS by strengthening the perception and comprehension stages of situational awareness. It enables cyber-physical data to be contextualized spatially, supports scenario-driven exploration of system behavior, and provides a flexible foundation for future extensions, including richer data sources, higher positional accuracy, and more advanced simulation use cases.





4 Conclusions

This deliverable has identified and structured the technical enablers for perception and comprehension required to operationalize the technological requirements defined in D3.1 across heterogeneous and security-critical cyber-physical systems. By translating abstract requirements into concrete perception, comprehension, and cross-cutting enablers, D3.2 establishes a solid technical foundation for achieving secure and reliable situational awareness.

The defined perception enables trustworthy acquisition of heterogeneous sensor and contextual data, supporting accurate position awareness, real-time environment monitoring, and secure interaction between physical and digital system components. In parallel, comprehension enablers provide mechanisms for combining, contextualizing, and interpreting perceived data to derive meaningful situational understanding, including detection of abnormal conditions and assessment of system states. Cross-cutting enablers, such as secure data management, low-latency communication, semantic structuring, and digital twin integration, ensure consistency, integrity, and resilience across distributed CPS architectures.

5 References

All the references have been provided as footnotes is this document.